



Thakur Shree DPS College of Engineering & Management, Vasai (East)

First Year Engineering

Academic Year: 2026-2027

Assignment No: 02

Subject: BSC102/AP

Date: 07/10/2026

Max Marks: 10

Formula Reference:

- **Critical Angle for TIR:** $\theta_c = \arcsin\left(\frac{n_2}{n_1}\right)$ ($n_1 > n_2$)
 - **Fractional Refractive Index Change:** $\Delta = \frac{n_1 - n_2}{n_1}$
 - **Numerical Aperture:** $NA = \sqrt{n_1^2 - n_2^2} \approx n_1 \sqrt{2\Delta}$
 - **Acceptance Angle:** $\theta_a = \arcsin\left(\frac{NA}{n_0}\right)$
 - **Attenuation Coefficient:** $\alpha \text{ [dB/km]} = \frac{10}{L} \log_{10}\left(\frac{P_{in}}{P_{out}}\right)$
-

Problem 1: Critical Angle for TIR and Effect of Ambient Medium

A step-index glass fiber has a core refractive index $n_1 = 1.54$ and a cladding refractive index $n_2 = 1.48$.

- Calculate the critical angle (θ_c) at the core-cladding interface.
- If a portion of the cladding is stripped away and the unclad core is exposed directly to ambient water ($n_{\text{water}} = 1.333$), calculate the new critical angle (θ'_c) at the core-water interface.
- What happens to the minimum angle of incidence required inside the core to maintain total internal reflection when the surrounding medium shifts from cladding glass to water?

Given: $n_1 = 1.54$, $n_2 = 1.48$, $n_{\text{water}} = 1.333$

(a) Critical angle at core-cladding interface:

$$\sin \theta_c = \frac{n_2}{n_1} = \frac{1.48}{1.54} \approx 0.9610 \implies \theta_c = \arcsin(0.9610) \approx 73.95^\circ$$

(b) Critical angle at core-water interface:

$$\sin \theta'_c = \frac{n_{\text{water}}}{n_1} = \frac{1.333}{1.54} \approx 0.8656 \implies \theta'_c = \arcsin(0.8656) \approx 59.95^\circ$$

Problem 2: Fractional Refractive Index Change and Cladding Index Determination

An optical fiber is manufactured with a core refractive index of $n_1 = 1.475$ and is designed to have a fractional refractive index change of $\Delta = 1.35\% = 0.0135$.

- (a) Determine the exact refractive index of the cladding (n_2).
- (b) Determine the Numerical Aperture (NA) using both the exact formula and the approximate formula $n_1\sqrt{2\Delta}$, and compute the percentage error between the two values.

Given: $n_1 = 1.475$, $\Delta = 0.0135$

(a) Cladding refractive index n_2 :

$$\Delta = \frac{n_1 - n_2}{n_1} \implies n_2 = n_1(1 - \Delta) = 1.475 \times (1 - 0.0135) = 1.475 \times 0.9865 \approx 1.4551$$

(b) Numerical Aperture comparison:

- *Exact NA:*

$$\text{NA}_{\text{exact}} = \sqrt{n_1^2 - n_2^2} = \sqrt{(1.475)^2 - (1.4551)^2} = \sqrt{0.0583} \approx 0.2415$$

- *Approximate NA:*

$$\text{NA}_{\text{approx}} = n_1\sqrt{2\Delta} = 1.475 \times \sqrt{2 \times 0.0135} = 1.475 \times \sqrt{0.0270} \approx 0.2424$$

- *Percentage discrepancy:*

$$\% \text{ Error} = \frac{|\text{NA}_{\text{approx}} - \text{NA}_{\text{exact}}|}{\text{NA}_{\text{exact}}} \times 100\% = \frac{0.2424 - 0.2415}{0.2415} \times 100\% \approx 0.37\%$$

Problem 3: Numerical Aperture, Acceptance Angle, and Medium Transition

A step-index fiber has a core index $n_1 = 1.50$ and a cladding index $n_2 = 1.45$.

- (a) Calculate the Numerical Aperture (NA) of the fiber.
- (b) Find the maximum acceptance angle (θ_a) when light is coupled into the fiber core from air ($n_0 = 1.00$).
- (c) Calculate the new acceptance angle (θ'_a) if the fiber entrance terminal is submerged in ethyl alcohol ($n'_0 = 1.36$).
- (d) Calculate the solid acceptance angle ($\Omega \approx \pi \sin^2 \theta_a$) in air.

Given: $n_1 = 1.50$, $n_2 = 1.45$

(a) Acceptance angle in air ($n_0 = 1.00$):

$$\sin \theta_a = \frac{\text{NA}}{n_0} = 0.3841 \implies \theta_a = \arcsin(0.3841) \approx 22.59^\circ$$

(b) Acceptance angle in ethyl alcohol ($n'_0 = 1.36$):

$$\sin \theta'_a = \frac{\text{NA}}{n'_0} = \frac{0.3841}{1.36} \approx 0.2824 \implies \theta'_a = \arcsin(0.2824) \approx 16.40^\circ$$

Problem 4: Attenuation Coefficient: Output Power and Link Length

An optical signal of power $P_{\text{in}} = 2.5 \text{ mW}$ is coupled into a silica fiber link of length $L = 12 \text{ km}$. The output power detected at the far end of the link is measured to be $P_{\text{out}} = 125 \text{ }\mu\text{W}$.

- (a) Calculate the overall attenuation (loss) in decibels (dB) across the link.
- (b) Determine the attenuation coefficient (α) of the fiber in dB/km.
- (c) If the minimum detectable power at the receiver is $P_{\text{min}} = 10 \text{ }\mu\text{W}$, determine the maximum transmission distance (L_{max}) possible before requiring a repeater.

Given:

$$P_{\text{in}} = 2.5 \text{ mW} = 2500 \text{ }\mu\text{W}, \quad P_{\text{out}} = 125 \text{ }\mu\text{W}, \quad L = 12 \text{ km}$$

(a) Total attenuation in dB:

$$\text{Loss}_{\text{dB}} = 10 \log_{10} \left(\frac{P_{\text{in}}}{P_{\text{out}}} \right) = 10 \log_{10} \left(\frac{2500}{125} \right) = 10 \log_{10}(20) \approx 10 \times 1.301 = 13.01 \text{ dB}$$

(b) Attenuation coefficient α :

$$\alpha = \frac{\text{Loss}_{\text{dB}}}{L} = \frac{13.01 \text{ dB}}{12 \text{ km}} \approx 1.084 \text{ dB/km}$$

(c) Maximum transmission distance (L_{max}): Maximum allowable total loss:

$$\text{Loss}_{\text{max}} = 10 \log_{10} \left(\frac{P_{\text{in}}}{P_{\text{min}}} \right) = 10 \log_{10} \left(\frac{2500 \text{ }\mu\text{W}}{10 \text{ }\mu\text{W}} \right) = 10 \log_{10}(250) = 23.98 \text{ dB}$$

$$L_{\text{max}} = \frac{\text{Loss}_{\text{max}}}{\alpha} = \frac{23.98 \text{ dB}}{1.084 \text{ dB/km}} \approx 22.12 \text{ km}$$